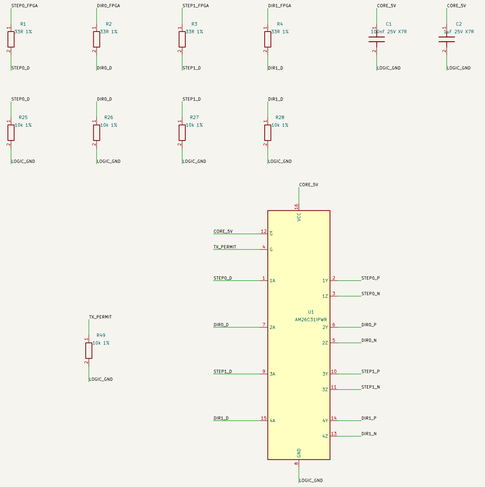
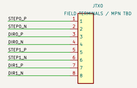
TWELVE-MOTOR STEP/DIR TRANSMITTER CANDIDATE

24 differential signals / 6 AM26C31IPWR banks. New electrical scope; current firmware remains six-channel.

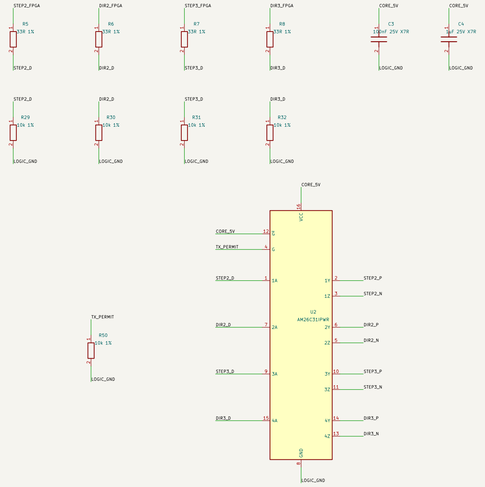
BANK TX0 / U1: JOINTS 0 AND 1



Place C1 close to U1 pins 16/8.
JTX0 matches allocation positions 1-8: reference is separate.



BANK TX1 / U2: JOINTS 2 AND 3

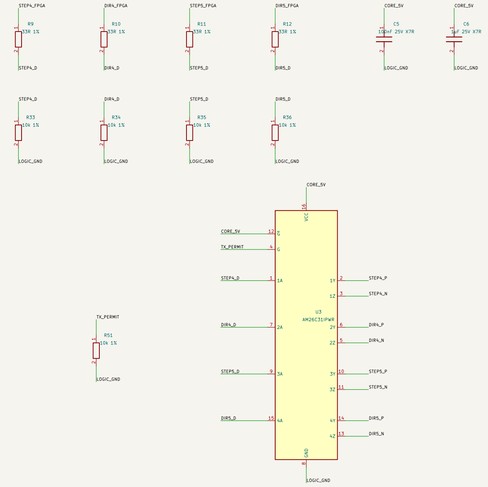


Place C3 close to U2 pins 16/8.
JTX1 matches allocation positions 1-8: reference is separate.



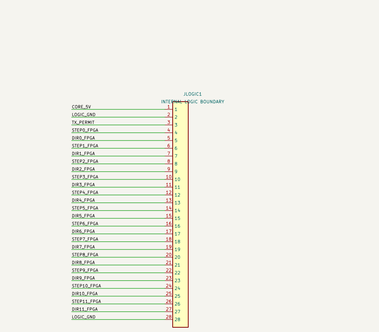
BANK TX2 / U3: JOINTS 4 AND 5

Engineering circuit only.
All twelve channels populated.
8-channel population needs
separate BOM and export.
No price or speed approval.

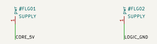


Place C5 close to U3 pins 16/8.
JTX2 matches allocation positions 1-8: reference is separate.

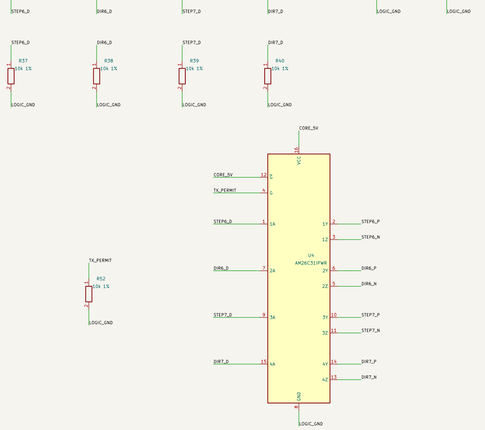








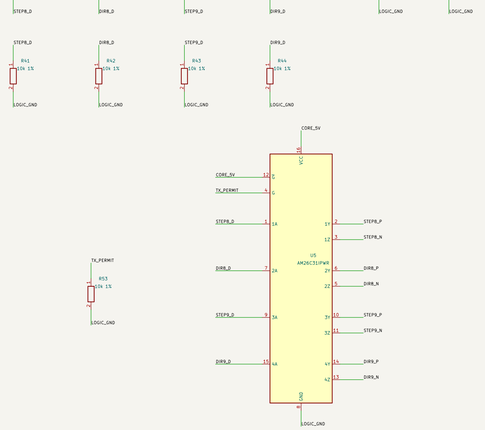
BANK TX3 / U4: JOINTS 6 AND 7



Place C7 close to U4 pins 16/8.
JTX3 matches allocation positions 1-8: reference is separate.



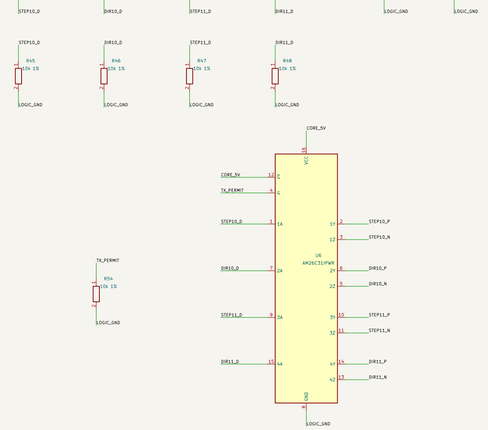
BANK TX4 / U5: JOINTS 8 AND 9



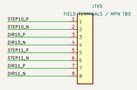
Place C9 close to U5 pins 16/8.
JTX4 matches allocation positions 1-8: reference is separate.



BANK TX5 / U6: JOINTS 10 AND 11



Place C11 close to U6 pins 16/8.
JTX5 matches allocation positions 1-8: reference is separate.





TX_PERMIT low makes differential outputs HIGH-Z. A remote drive may interpret floating lines incorrectly; receiver bias and independent drive inhibit need qualification.
1.2A reservation on CORE_5V is a conservative design allowance, not a measured load. Existing six-channel converter budget cannot be reused without recalculation.
Twelve motor channels are not twelve Cartesian axes. No gateware expansion, cable qualification, thermal proof or manufacturing release follows from this drawing.

Source: TI AM26C31 SLLS103P, pin Table4-1, operating Section5.3, electrical Section5.5, function Table7-1.
Input33R damping,10k defaults, capacitor MPNs, connector selections and final protection remain open.

Partial schematic only; not released for manufacture

Sheet: /	
File: tx_twelve.kicad_sch	
Title: Twelve-channel differential transmitter candidate	
Size: A1	Date: 2026-09-28
KiCad E.D.A. 10.0.4	**Rev: r0.2**
	Id: 1/1